

# Sanjit Singh

AR/VR Developer

Email: [sanjitpsingh@yahoo.com](mailto:sanjitpsingh@yahoo.com)

Cell Phone: 609-529-2993

LinkedIn: <https://www.linkedin.com/in/sanjit-singh-66374b6a/>

Portfolio: <https://scarletknight11.github.io/>

YT: <https://www.youtube.com/@sanjitsingh3693/videos>

Experienced Unity & AI Developer with over 8 years of expertise in developing and optimizing 3D simulations and interactive environments across mobile, desktop, and VR platforms. Proficient in C# and Unity scripting, with a strong background in creating generative models, AI-driven interactions, and seamless UI/UX integration. Adept at collaborating with cross-functional teams to translate architectural and technical specifications into high-performance, scalable simulations.

## **SKILLS**

**Programming Languages:** C# (Unity), Python, Java, C/C++, JavaScript, Swift, MATLAB

**Platforms:** Unity3D, Unreal Engine 4/5, Android Studio, Visual Studio, Oculus, HoloLens, Smart Glasses

**Tools/Technologies:** 3D Simulation, AI Behavior Implementation, UI/UX Development, Performance Optimization, Unity ML-Agents Toolkit, Image Processing, VR/AR Development, ARCore, ARFoundation, Pixyz SDK, Procedural Generation, Post Processing, Revit (Familiarity), NumPy, Pandas, OpenCV

**Mobile Development:** Android, iOS (Game Development)

**Additional Skills:** Architectural Visualization, Generative Modeling, Multiplayer Support, Voice Command Integration, Cloud-Based Asset Management

## **EDUCATION**

Rutgers University, New Brunswick, NJ

September 2020 – December 2021

Bachelor of Science in Computer Science

New Jersey Institute of Technology, Newark, NJ

September 2019 – May 2020

Transferred to Rutgers University - New Brunswick.

Mercer County Community College, West Windsor, NJ

Graduated May 2019

Associates of Science in Computer Science

Recognized in school article: [http://www.mccc.edu/news/2019/general/Olu\\_Sanjit\\_Internships.shtm](http://www.mccc.edu/news/2019/general/Olu_Sanjit_Internships.shtm)

## **Teaching Experience**

Outschool

January 2026 - Present

*Instructor*

## Code Wiz

April 2024 – September 2024

### *Coding & Robotics Coach*

Taught courses in Unity3D & Roblox to bunch of kids ages from 7-17.

## Mercer County Community College

August 2023 – May 2024

### *Dream Mentor*

Worked as a mentor/tutor with neurodiverse students that have intellectual disabilities in subjects such as Mathematics, Information Systems Technology, Computer Science, .NET, Game Design & Digital Media.

Parts of responsibility involved peer mentoring, note taking for students during classroom lectures, tutoring on homework assignments & exam preparation.

## **WORK EXPERIENCE**

## EliteCAD Designs

October 2023 – February 2026

### *Computer Vision Engineer*

Developing a multi-device AI and computer vision platform for building operations using ESP32-CAM microcontrollers, integrating Arduino IDE, Python, and OpenCV to process real-time video streams and transcribe audio for collaborative, multi-user maintenance workflows.

Designing an audio transcription and data pipeline that converts spoken operational data into structured text and securely syncs it to Firebase, enabling centralized maintenance tracking and control across distributed teams.

Building a smart-glasses–based workflow using Vuzix Blade 2, leveraging Android Studio to capture audio, perform speech-to-text transcription, and stream operational insights directly to a cloud backend in real time.

Implementing computer vision–based object detection for building operations using OpenCV and YOLO, with custom dataset creation and training in Python utilizing NumPy, TensorFlow, and Scikit-Learn, then deploying trained models to Android for real-time detection on smart glasses.

Creating an equipment identification system by training and deploying custom object-detection models using YOLO and Roboflow, enabling accurate recognition of building infrastructure and maintenance assets in live operational environments.

Developing an AI chatbot web interface in Python using LangChain, Pandas, OpenAI APIs, and Streamlit to support intelligent querying, documentation lookup, and decision support for building operations teams.

Integrating Omi wearable AI devices to transcribe in-field conversations and stream structured transcripts to a webhook-based backend, enhancing situational awareness and post-operation analysis.

Building an XR authoring and visualization platform in Unreal Engine 5, using Blueprints and C++ to capture

3D Gaussian Splats of buildings and store them in a cloud-based system for remote access, visualization, and collaboration across clients.

Valkyra Technologies

June 2024 – August 2025

*Co-Founder & CTO*

Early-stage startup focusing on leveraging augmented reality technology & medical robotics utilizing machine learning & computer vision for emergency medicine with specific focus on trauma care for military soldiers in rural settings that were brutally injured during combat battlefield.

Mount Sinai Innovation Partners

September 2024 – December 2024

*Innovation Fellow*

MSIP's Fellows Program is designed to educate and engage early-career scientists, attorneys, and/or business professionals who have strong interests in the commercialization process of new therapeutics, diagnostics, medical devices or digital health technologies to the worldwide patient population. If successfully completed, the fellow will be prepared to leverage their expertise in science, business and law to help drive research discoveries to the marketplace. The two-phase scheme will train participants in the various aspects of technology commercialization.

Imagining Reality Insights & Solutions

December 2023 – August 2024

*Lead Programmer*

- NSF I-Corps funded startup from the University of Maryland, College Park that focuses on developing virtual reality games for special education students with dyslexia using Unity3D.  
Led a team of developers in creating VR games tailored for special education students with dyslexia, leveraging Unity3D to ensure seamless performance on mobile platforms.
- Designed and implemented advanced user interfaces, focusing on optimizing responsiveness and delivering a pixel-perfect user experience across various devices.
- Spearheaded weekly code reviews, offering detailed feedback and mentoring junior developers to maintain high coding standards.
- Developed and integrated experimental XR technologies, creating reusable assets and game mechanics that were utilized across multiple projects.
- Pioneered the use of ManoMotion Unity SDK for simulating mobile hand tracking, enhancing the immersive experience in mobile AR games.
- both HoloLens and mobile devices.

Massachusetts General Brigham Women's Hospital/Harvard Medical School

September 2023 – Jan 2024

*Research Intern*

- Collaborated on a groundbreaking AR project for trauma care, focusing on real-time telemonitoring systems using Unity3D and HoloLens 2.
- Developed complex UI panel controls for vital monitoring and checklist instructions, enhancing the usability of the system in high-pressure emergency scenarios.
- Successfully integrated physiological data and video streaming SDKs, enabling real-time monitoring and collaboration across multiple platforms.
- Authored comprehensive technical documentation to guide clinicians in the effective use of the AR system during trauma care procedures.

Johns Hopkins University

August 2021 – June 2023

*Research Collaborator*

- Created VR applications using Unity3D for glaucoma patients, focusing on enhancing visual field tests and improving clinical outcomes.
- Engineered visual renderings for VR environments, utilizing advanced image processing techniques in MATLAB to fine-tune contrast levels and object recognition.
- Conducted detailed data analysis to assess the effectiveness of VR interventions, providing insights that informed subsequent research and development.

Hoth Intelligence

August 2022 – December 2022

*XR Contractor*

- Led the development of AR technologies for real-time medical procedures in skull surgery, utilizing Unity3D and HoloLens 2.
- Implemented Unity's Barracuda Neural Network framework to achieve precise face mesh tracking, significantly improving the accuracy of medical procedures.
- Enhanced AR overlay tracking for catheter procedures by integrating Vuforia SDK and MRTK, achieving superior tracking efficiency and reliability.

UMD Applied Research Lab for Intelligence & Security

June 2022 – August 2022

*RISC Intern*

- Conducted in-depth analysis of population distribution models using open-source instrumentation libraries in Python, contributing to strategic decision-making processes in intelligence and security.
- Installed and configured an Apache server through Docker, enabling the processing of large datasets for population analysis.
- Utilized Prophet for forecasting and NumPy, Pandas, and Matplotlib libraries in JupyterLabs to visualize and predict population trends based on CSV data files.
- Implemented KMeans clustering in Python to classify and sort data, aiding in the analysis of complex datasets.

MIT

Jan 2022 – July 2022

*Catalyst Fellow*

- Selected as the first undergraduate student in the prestigious Catalyst Fellowship at MIT, where I collaborated with multidisciplinary teams to identify and address unmet medical needs.
- Proposed and developed a novel solution for stroke rehabilitation focused on the upper arm extremity, presenting weekly updates to faculty with comprehensive research and evidence.
- Participated in stakeholder and customer discovery interviews to gain insights into the challenges faced by patients and healthcare providers, refining the project's focus.
- Collaborated with experts from various fields to create a robust action plan for the development and commercialization of the proposed solution.

*S.U.I.T.S Intern*

- Supported the development of AR systems for astronaut training as part of the NASA SUITS program, utilizing Unity3D and the Microsoft HoloLens for HUD design and telemetry streaming.
- Contributed to the technical challenges of future NASA missions by experimenting with and documenting the use of AR technologies in complex space scenarios.
- Played a key role in ensuring the technical feasibility of AR solutions, troubleshooting and resolving issues in real-time during development, testing, and deployment.
- Assisted in promoting the NASA SUITS challenge, helping to design and implement promotional campaigns to increase participation and awareness.

MPLEX

August 2021 – Jan 2022

*Mixed Reality Developer*

- Developed innovative AR Spectator app features that allowed users to replay gameplay events in real-time, enhancing user engagement and interaction.
- Modified and optimized features such as collision detection, laser hit detection, and UI elements in Unity3D to improve the app's performance and user experience.
- Contributed to the successful Epic Games Mega Grant proposal by providing detailed revisions and ensuring the accuracy of the grant application, securing funding for continued development.
- Integrated the Twitch API into Unity, enabling live chat streaming through the AR app, which enhanced the interactive experience for users.

Rutgers University @ Computational Biomedicine Imaging and Modeling Center

August 2021 – Dec 2021

*Research Assistant @ The Intelligent Visual Interface Laboratory*

- Conducted advanced AI research focused on developing intelligent models for multi-agent systems within Unity3D, specifically for medical and sports simulations.
- Implemented machine learning algorithms using Unity's ML-Agents Toolkit to enhance AI decisionmaking and behavioral logic in multi-agent environments.
- Utilized AI-driven pathfinding and decision trees to create realistic and responsive interactions between AI agents in simulation environments, improving the overall accuracy and realism of the models.
- Applied reinforcement learning techniques to train AI agents to perform specific tasks within simulated environments, leading to more adaptive and efficient behaviors.
- Collaborated with interdisciplinary teams to integrate AI solutions into broader simulation frameworks, contributing to published research and academic presentations.

The University of Auckland

May 2021 – September 2021

*Research Intern @ Empathic Computing Lab*

- Conducted pioneering research on the interaction between eye-gaze tracking, inter-brain synchrony, and collaborative virtual reality, especially in the context of online counseling during the COVID-19 period.

- Analyzed brain hyper-scanning data using Python libraries such as HyPyP and MNE, providing insights into communication patterns and emotional interactions in virtual environments.
- Developed and analyzed realistic virtual worlds in VR to better understand brain synchronization and interpersonal communication, contributing to the lab's ongoing research into empathetic computing.
- Utilized SciPy, NumPy, and Pandas to process and analyze eye-tracking data, enabling the creation of more immersive and emotionally resonant virtual environments.

#### National Security Innovation Network

June 2021 – August 2021

##### *X-Force Fellow – Software Engineering Intern*

- Utilized storyboarding techniques to showcase future technologies in a video/VR format, contributing to the development of innovative training and planning tools for military applications.
- Designed and implemented immersive scenarios for warfighter jet training and combat simulation using Unreal Engine 4, enhancing the realism and effectiveness of military training programs.
- Developed blueprint scripts for virtual characters, HUD, and AI agents, ensuring that the simulations provided a high level of detail and interactivity.
- Advocated for the investment in modern storytelling tools, helping to secure funding and support for the development of these technologies within the military sector.

#### NASA SUITS Augmented Reality Challenge

September 2020 – May 2021

##### *Project Team Leader*

- Led a top 20 team in the NASA SUITS Augmented Reality challenge, focusing on the development of AR systems to assist astronauts during lunar exploration missions.
- Utilized the Microsoft Mixed Reality Toolkit (MRTK) to develop UI components, gesture recognition features, and voice commands for the Heads-Up Display (HUD) on the HoloLens 2.
- Implemented telemetry streaming using Postman API and Unity's JSON package, enabling real-time monitoring of space vitals and other critical data during missions.
- Developed geology sampling documentation UI that provided astronauts with accurate information and guidance on what to collect as they navigated the lunar surface.
- Worked on implementing a navigation component to assist astronauts in planning waypoints and routes on the moon, contributing to the overall mission success.
- Collaborated with the University of Washington Reality Lab to further develop and refine the project, with plans to submit it to the NASA SUITS challenge for the 2022-2023 cycle.

#### XBoost

February 2020 – April 2021

##### *Co-founder & Vice President*

- Co-founded an early-stage biotechnology startup funded by the NSF I-Corps program, focusing on the development of rehabilitative assistive robotic exoskeleton devices and VR/AR environments for medical applications.
- Managed a cross-disciplinary team of computer scientists, software engineers, hardware engineers, biomedical engineers, and medical clinicians, driving the project from concept to prototype.

- Developed VR/AR interactive environments tailored for patients with stroke, musculoskeletal conditions, and orthopedic rehabilitation, enhancing their recovery through innovative technology.
- Played a key role in securing pending collaborations with Mount Sinai Hospital and the University of Pennsylvania Robotics Rehabilitation Lab, positioning XBoost as a leader in the field of medical technology.
- Despite challenges posed by COVID-19, contributed to the project's progress and laid the groundwork for future development and commercialization.

#### New Jersey Institute of Technology

August 2019 – September 2020

#### *Research Assistant @ Biodynamics & MIXR Lab*

##### Biodynamics Lab

- **Virtual Rehabilitation Therapy Research:** Conducted extensive research on the development of robotic exoskeleton arms designed for neurological patients with cerebral palsy. Focused on creating solutions that provide neuromuscular support to improve the quality of life for these patients.
- **Hardware and Software Integration:** Integrated advanced hardware components, including Arduino microcontrollers, IMU sensors, and Myoware muscle sensors, with Unity3D to develop a wearable controller system. This system enabled patients to interact with therapeutic video games through natural movements, facilitating rehabilitation exercises.
- **Prototyping and Development:** Led the design and prototyping of the exoskeleton arm, ensuring that the device was both functional and comfortable for patients. Developed C# scripts in Unity3D to translate sensor data into responsive in-game actions, creating an immersive and effective therapy tool.
- **NSF I-Corps Program:** Played a key role in securing \$2,000 in funding through the NSF I-Corps program, which supported the commercialization of the lab's research. Developed a comprehensive business plan and explored potential partnerships to bring the exoskeleton technology to market.
- **Collaboration and Documentation:** Collaborated with biomedical engineers, clinicians, and software developers to refine the exoskeleton system. Documented research findings, technical specifications, and user feedback, contributing to academic publications and presentations at conferences.

##### MIXR Lab

August 2019 – January 2020

- **Military Medicine Applications:** Conducted research focused on the development of advanced tools for paramedics and surgeons in military settings, leveraging Unity3D and cutting-edge machine learning and computer vision technologies.
- **Motion Tracking and 3D Mesh Scanning:** Utilized Azure Kinect DK for precise motion tracking and 3D mesh scanning, enabling accurate visualization of a patient's body in a simulated environment. This technology was aimed at assisting medical professionals in diagnosing and treating injuries on the battlefield.
- **Unity3D and Machine Learning:** Integrated machine learning algorithms within Unity3D to develop tools that could assess and predict injury outcomes based on motion and visual data. Implemented real-time data analysis to provide immediate feedback to medical teams during simulations.
- **Pose Estimation and Body Mapping:** Developed algorithms for accurate pose estimation and body mapping using the Azure Kinect SDK. These algorithms were crucial for creating realistic and interactive simulations that could be used in training and real-world medical scenarios.
- **Holographic Visualization:** Explored the use of holographic projections with the Microsoft HoloLens 2 to overlay 3D models of human anatomy onto a patient's body. This innovation was designed to

assist surgeons in visualizing the internal structure of injured soldiers, improving the accuracy and effectiveness of surgical interventions.

- **Research Outcomes and Dissemination:** Contributed to the lab's research output by co-authoring papers and presenting findings at military medicine and technology conferences. The work was aimed at enhancing the capabilities of field medics and surgeons in austere environments.

Enable Games / Drexel University Spin off startup@ Replay Laboratory

May 2018 – June 2020

*Game Developer & Researcher*

- Developed Unity3D-based games aimed at enhancing physical therapy for patients with neurological conditions such as cerebral palsy and Parkinson's disease.
- Built a custom asset library that integrated Xbox Kinect for motion tracking, enabling patients to engage in therapeutic activities through interactive games.
- Utilized Stykz motion capture studio software to create realistic animations based on real-life movements, contributing to the effectiveness of the games in clinical settings.
- Analyzed clinical trial data to assess the usability and effectiveness of the games, providing insights that informed the design and development of future therapeutic interventions.

Weill Cornell Medicine / Cornell Medical School

April 2019 – June 2019

*Research Assistant*

- Worked on an engineering sub-team researching the integration of 3D segmented heart models with fluoroscopic images to improve the accuracy of catheter placement during cardiovascular interventions.
- Utilized AR technologies to visualize and minimize errors in syncing CAT scan and fluoroscopic images, enhancing the precision and safety of the procedures.
- Improved the thickness of blood volume in 3D segmented CAD models using Rhino 3D and Geomagic Wrap, testing the visualizations on the Microsoft HoloLens for real-time application in the operating room.
- collaborated with interventional cardiologists to refine the AR system, contributing to the development of a cutting-edge tool for cardiovascular surgery.

## **HACKATHON PROJECTS**

OperatAR

Dec 2017 - Present

An app that teaches users how to perform surgery and how to give clear directions in an operation room through Augmented Reality. The app was built using Unity3D, C#, Vuforia and Android Studio integrated through Google Cloud using speech to text API.

CerebralPalsyVR

Nov 2017 – Nov 2017

A series of VR-based games that help rehabilitate cerebral palsy patients who are partially able to move their feet and/or hands. The patient uses an Oculus Rift headset with Arduino sensors attached to their legs and hands. I worked extensively on the implementation of the main menu, UI, VR input. Also helped develop certain movements that our VR game could use to help cerebral palsy patients. Worked on backend with

Android Studio to use the gyroscope tool for our balance measurements with mobile Android devices serving as the sensors of physical motion.

## **AWARDS/HONOR**

### MIT LinQ Catalyst Fellowship

MIT LinQ Fall 2021

Selected as a Catalyst fellow for the 2022 cohort. Catalyst brings together multidisciplinary experts to work together in an iterative process to identify and validate unmet medical and health-related needs, discover new project opportunities, and develop action plans.

### Amazon Web Services - Best Use of AWS

HackRU Fall 2017

Developed a machine learning program that can diagnose Diabetes based on the information that patients provide to Amazon Alexa through a series of questions.

### Top 30 Hack

PennApps Fall 2018

Developed a social media app using Facial Recognition to help others get to know each person faster. The app scans their face and through the datasets display their name and stores the exact day, time, and location where they met.

### NSF I-Corps Grant Recipient

National Science Foundation Innovation Corps Spring 2020

\$2,000 grant awarded to "teams interested in exploring the commercial viability of their ideas for products and businesses that are based on their own inventions.

### MIT COVID-19 Challenge Winner

MIT COVID-19 Challenge Spring 2020

Came up with a program that can-do questionnaire testing and then based on answers get a sample test kit sent home. Once that test kit is sent back to clinical labs, they can use Machine Learning Algorithms to conduct antibody testing to accurately diagnose if the patient has Coronavirus or not.

## **Publications**

### [Eye-Gaze, Inter-Brain Synchrony, and Collaborative VR in Conjunction with online Counselling: A Pilot Study](#)

Ihshan Gumilar, Amit Barde, Ashkan F. Hayat, Mark Billingham, & Sanjit Singh

### [Inter-brain Synchrony and Eye Gaze Direction During Collaboration in VR](#)

Ihshan Gumilar, Amit Barde, Prasanth Sasikumar, Mark Billingham, Ashkan F. Hayati, Gun Lee, Yuda Munarko, Sanjit Singh, Abdul Momin